

This document is a **transcript** of the lecture, with extra summary and vocabulary sections for your convenience. Due to time constraints, the notes may sometimes only contain limited illustrations, proofs, and examples; for a more thorough discussion of the course content, **consult the textbook**.

Summary

Quick summary of today's notes. Lecture starts on next page.

- A function $f : X \rightarrow Y$ is *invertible* if it is both one-to-one and onto.
- The *identity function* on a set X is the function $\text{id}_X : X \rightarrow X$ with $\text{id}_X(x) = x$ for all $x \in X$.
If $f : X \rightarrow Y$ is a function then $f \circ \text{id}_X = \text{id}_Y \circ f = f$.
- A function $f : X \rightarrow Y$ is invertible if and only if there is a function $g : Y \rightarrow X$ such that $f \circ g = \text{id}_Y$ and $g \circ f = \text{id}_X$. If such a function g exists, then it is unique and we call it the *inverse* of f .
If we know that $f : X \rightarrow Y$ is invertible, then we write $f^{-1} : Y \rightarrow X$ to denote its inverse.
- Let m and n be positive integers.

The identity function on the set $\mathbb{R}^n$ is linear. Its standard matrix is the $n \times n$ *identity matrix*

$$I_n = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}.$$

- Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear. Then T is invertible only if $m = n$.
Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible. Then its inverse $T^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is also linear.
- A matrix is *invertible* if it is the standard matrix of an invertible linear transformation.
A matrix can be invertible only if it is square, i.e., has the same number of rows and columns.
- Suppose A is invertible. Then A is the standard matrix of an invertible linear transformation T .
We define the *inverse* A^{-1} of A to be the standard matrix of the linear transformation T^{-1} .
If A is $n \times n$ then A^{-1} is also $n \times n$, and $AA^{-1} = A^{-1}A = I_n$.
- A square matrix A is invertible if and only if $\text{RREF}(A) = I_n$ for some n .
- We can check if an $n \times n$ matrix A is invertible and try to compute A^{-1} at the same time.
We do this as follows. First form the $n \times 2n$ matrix $\begin{bmatrix} A & I_n \end{bmatrix}$.
Then compute the reduced echelon form of this matrix.
If $\text{RREF}(\begin{bmatrix} A & I_n \end{bmatrix}) = \begin{bmatrix} I_n & B \end{bmatrix}$ for some $n \times n$ matrix B , then $A^{-1} = B$.
Otherwise, A is not invertible.
- A 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if and only if $ad - bc \neq 0$, in which case

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

1 Last time: adding and multiplying matrices

Suppose $T : \mathbb{R}^r \rightarrow \mathbb{R}^s$ and $U : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are linear transformations with standard matrices A and B .

Let $c \in \mathbb{R}$ be a scalar.

1. The scalar multiple $cT : \mathbb{R}^r \rightarrow \mathbb{R}^s$ of T is the linear transformation with $(cT)(x) = cT(x)$.

The standard matrix of cT is cA where cA is given by multiplying every entry in A by c , i.e.:

Definition. cA is the matrix $\begin{bmatrix} ca_1 & ca_2 & \dots & ca_r \end{bmatrix}$ where $A = \begin{bmatrix} a_1 & a_2 & \dots & a_r \end{bmatrix}$.

2. If T and U have the **same domain and codomain**, meaning that $r = n$ and $s = m$ and that A and B have the same size, then we can form the sum $T + U : \mathbb{R}^n \rightarrow \mathbb{R}^m$ as the linear transformation with $(T + U)(x) = T(x) + U(x)$.

The standard matrix of $T + U$ is $A + B$ where:

Definition. $A + B$ is the matrix whose i th column is the sum of the i th columns of A and B .

3. If the domain of T is the codomain of U , meaning that $r = m$, then we can form the composition $T \circ U : \mathbb{R}^n \rightarrow \mathbb{R}^s$ as the linear transformation with $(T \circ U)(x) = T(U(x))$.

The standard matrix of $T \circ U$ is the product AB where:

Definition. The product AB of matrices A and B , where the number of columns of A is the number of rows of B , is the matrix $AB = \begin{bmatrix} Ab_1 & Ab_2 & \dots & Ab_n \end{bmatrix}$ where $B = \begin{bmatrix} b_1 & b_2 & \dots & b_n \end{bmatrix}$.

Some remarks.

$A + B$ is only defined if A and B are matrices of the same size.

We have $A + B = B + A$ when either side is defined.

AB is only defined if the number of columns of A is equal to the number of rows of B .

When defined, AB has the same number of rows as A , and the same number of columns as B .

Even if AB and BA are both defined, we may still have $AB \neq BA$.

Example. Let $A = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \end{bmatrix}$ be a 3×4 matrix.

Consider what happens when we multiply A on the left by various 3×3 matrices.

1. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} A = \begin{bmatrix} a & b & c & d \\ i & j & k & l \\ e & f & g & h \end{bmatrix}$. Multiplication swaps rows 2 and 3 of A .
2. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} a & b & c & d \\ 3e & 3f & 3g & 3h \\ i & j & k & l \end{bmatrix}$. Multiplication rescales row 2 by factor 3.
3. $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} a+2i & b+2j & c+2k & d+2l \\ e & f & g & h \\ i & j & k & l \end{bmatrix}$. This adds 2 times row 3 to row 1.

Matrix Multiplication:

$$\text{If } \mathbb{R}^n \xrightarrow{U} \mathbb{R}^m \xrightarrow{T} \mathbb{R}^k$$

$$T: \mathbb{R}^m \rightarrow \mathbb{R}^k \quad U: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

Outputs of U can be inputs to T .

so we can define $(T \circ U)(u) = T(U(u))$.

If T, U are linear with standard matrices A, B
 $(k \times m) \quad (m \times n)$

then define $AB \stackrel{\text{def}}{=} (\text{standard matrix of } T \circ U)$
 $\hookrightarrow$ is linear.

Example:

$$\begin{array}{ccc} \begin{bmatrix} 1 & 2 & 4 \\ 5 & 6 & 7 \end{bmatrix} & \begin{bmatrix} 3 & 5 \\ 4 & 4 \\ 1 & 2 \end{bmatrix} & = \begin{bmatrix} 3+8+4 & 5+8+8 \\ 15+24+7 & 25+24+14 \end{bmatrix} \\ \begin{matrix} (k \times m) \\ 2 \times 3 \end{matrix} & \begin{matrix} (m \times n) \\ 3 \times 2 \end{matrix} & \\ & & = \begin{bmatrix} 15 & 21 \\ 46 & 63 \end{bmatrix} \end{array}$$

$$AB = A \begin{bmatrix} \vec{b}_1 & \vec{b}_2 & \vec{b}_3 & \dots & \vec{b}_n \end{bmatrix} \stackrel{\text{def}}{=} \begin{bmatrix} A\vec{b}_1 & A\vec{b}_2 & A\vec{b}_3 & \dots & A\vec{b}_n \end{bmatrix} \quad \begin{matrix} 2 \times 2 \end{matrix}$$

- ① AB only defined when # of rows (B) = # of cols (A)
- ② Even when AB and BA are both defined you often have $AB \neq BA$.
- ③ Every row operation on A can be expressed as multiplication on the left by an elementary matrix. [Previous page last example].

Elementary matrix:

An elementary matrix is formed by doing one row operation to the identity matrix. [refer to next page].

$$I = \begin{bmatrix} 1 & & & 0 \\ & \ddots & & \\ 0 & & \ddots & 1 \end{bmatrix}$$

$$\begin{array}{ccc} \textcircled{1} & \textcircled{2} & \textcircled{3} \\ \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & 0 & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix} & \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & k & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix} & \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & a & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix} \end{array}$$

$$k \neq 0$$

★ What is the use of elementary matrices?

How does it connect to 3rd bullet on previous page.

Each of the matrices E that are left-multiplied with A are examples of *elementary matrices*, which are defined to be matrices formed by doing exactly one row operation to the *identity matrix*

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

that has all 1's on the diagonal and all 0's in off-diagonal positions.

If E is an elementary matrix, then EA is obtained from A by doing one row operation. (Check this!)

Moral: individual row operations on A correspond to multiplying A on the left by elementary matrices.

2 Matrix transpose

The *transpose* of an $m \times n$ matrix A is the $n \times m$ matrix $A^\top$ whose columns are the rows of A .

If a_{ij} is the entry in row i and column j of A , then this is the entry in row j and column i of $A^\top$.

If $A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$ and $A^\top = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$. If $C = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -3 & 5 & -2 & 7 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ then $C^\top = \begin{bmatrix} 1 & -3 & 0 \\ 1 & 5 & 0 \\ 1 & -2 & 1 \\ 1 & 7 & 0 \end{bmatrix}$.

The transpose of A is given by flipping A across the main diagonal, in order to interchange rows/columns.

Basic properties of the transpose operation:

- $(A^\top)^\top = A$ since flipping twice does nothing.
- If $c \in \mathbb{R}$, and A and B have the same size, then $(A + B)^\top = A^\top + B^\top$ and $(cA)^\top = c(A^\top)$.
- If A is an $k \times m$ matrix and B is an $m \times n$ matrix then $(AB)^\top = B^\top A^\top$.

↪ use formula and compute and then compare.

3 Inverses

Let $f : X \rightarrow Y$ be a function with domain X and codomain Y .

Definition. The function f is *invertible* or *bijective* if f is both onto and one-to-one.

Here is a more direct definition of an invertible function:

Proposition. f is invertible if and only if for each $b \in Y$ there is exactly one $a \in X$ with $f(a) = b$.

Proof. f is onto if and only if for each $b \in Y$ there is at least one input $a \in X$ with $f(a) = b$.

f is one-to-one if and only if for each $b \in Y$ there is at most one input $a \in X$ with $f(a) = b$.

Therefore f is both onto and one-to-one if and only if the given condition holds. □

The *identity function* on a set X is the function $\text{id}_X : X \rightarrow X$ with $\text{id}_X(a) = a$ for all $a \in X$.

If $f : X \rightarrow Y$ and $g : Y \rightarrow X$ are any functions then $f \circ \text{id}_X = f$ and $\text{id}_Y \circ g = g$.

$$f: X \rightarrow Y$$

Def ① f invertible $\iff f$ is both ONTO and ONE-TO-ONE

①

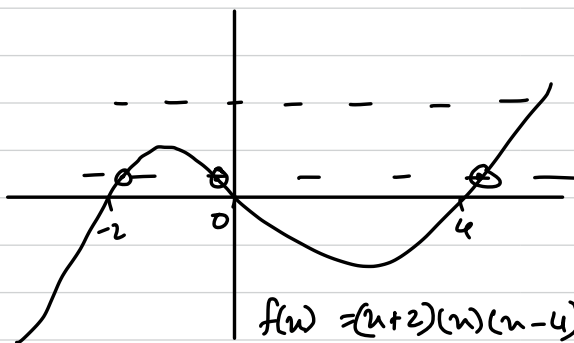
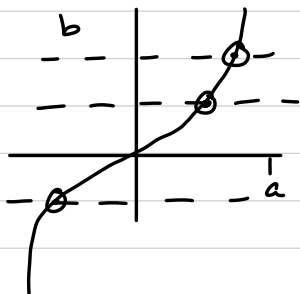
Def ② f invertible if for each $b \in Y$ there is exactly one $a \in X$ with $f(a) = b$.

Def ③ f invertible if it has an inverse which is a function $f^{-1}: Y \rightarrow X$ with $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$

Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$: f is invertible when every horizontal intersects its graph exactly once.

invertible $f(n) = n^3$

not invertible



$3 > 1$

Identity function (on set X) is $\text{id}_X: X \rightarrow X$

$$\text{id}_X(a) = a$$

Ex: $\text{id}_{\mathbb{R}^n}$ is linear w/ standard matrix

$$I = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix} = [e_1 \ e_2 \ e_3 \ \dots \ e_n]$$

if $f: X \rightarrow Y$ then $\text{id}_Y \circ f = f \circ \text{id}_X = f$

$\hookrightarrow$ what does this mean?

Example. The identity function on $\mathbb{R}^n$ is the linear transformation $\text{id}_{\mathbb{R}^n} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ whose standard matrix is the $n \times n$ *identity matrix*

$$I_n = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}.$$

If AI_n is defined then $AI_n = A$. If $I_n A$ is defined then $I_n A = A$.

Even more concretely, a function is invertible if and only if it has an *inverse* in the following sense.

Proposition. The function $f : X \rightarrow Y$ is invertible if and only if there is a function $f^{-1} : Y \rightarrow X$ such that $f \circ f^{-1} = \text{id}_Y$ and $f^{-1} \circ f = \text{id}_X$.

If there exists a function f^{-1} with these properties, then it is unique, and we call it the *inverse* of f .

Proof. We first prove that the inverse is unique, and then prove the first statement. This takes 3 steps.

1. Suppose $g : Y \rightarrow X$ and $h : Y \rightarrow X$ both have $f \circ g = f \circ h = \text{id}_Y$ and $g \circ f = h \circ f = \text{id}_X$.

Then $g \circ (f \circ h) = g \circ \text{id}_Y = g$ and $(g \circ f) \circ h = \text{id}_X \circ h = h$.

But $g \circ (f \circ h)$ and $(g \circ f) \circ h$ are the same: given any $b \in Y$, they both have output $g(f(h(b)))$.

Therefore $g = h$, so if f has an inverse, then it is unique.

2. If f has an inverse $f^{-1} : Y \rightarrow X$, then for each $b \in Y$ we have $f(a) = b$ for $a = f^{-1}(b)$.

This value of $a \in X$ is unique since if $f(a) = f(a') = b$ then

$$a = \text{id}_X(a) = (f^{-1} \circ f)(a) = f^{-1}(f(a)) = f^{-1}(f(a')) = (f^{-1} \circ f)(a') = \text{id}_X(a') = a'.$$

Therefore, by the previous proposition, if f has an inverse then f is invertible.

3. Suppose f is invertible. Define $f^{-1}(b)$ for $b \in Y$ to be the unique $a \in X$ such $f(a) = b$.

This defines a function $f^{-1} : Y \rightarrow X$ with $f(f^{-1}(b)) = f(a) = b$ and $f^{-1}(f(a)) = a$.

This is the same as saying $f \circ f^{-1} = \text{id}_Y$ and $f^{-1} \circ f = \text{id}_X$.

□

Example. Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the linear function $T(v) = Av$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

We have $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \text{RREF}(A)$.
 $\rightarrow T$ is onto as A has pivot in every row
 $\rightarrow T$ is one-to-one as A has pivot in every column.

This means A has a pivot position in every row and every column.

By results in previous lectures, we know that this implies that T is onto and one-to-one, i.e., bijective.

What is the inverse T^{-1} of T ?

Note that $T^{-1}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)$ is the unique vector $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ such that $Ax = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

We can solve for x by row reducing the augmented matrix of this matrix equation:

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3/2 \end{bmatrix},$$

Key observation. for any given $b \in \mathbb{R}^2$, we can solve for $u \in \mathbb{R}^2$
with $\underbrace{T(u)}_{Au} = b$

Ex: What x has $T(u) = Au = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\text{RREF}\left(\begin{bmatrix} A & 0 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3/2 \end{bmatrix} \Leftrightarrow \begin{cases} u_1 = -2 \\ u_2 = 3/2 \end{cases}$$

Ex: what u has $T(u) = Au = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
 $u = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}$

We have just computed $T^{-1}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 3/2 \end{bmatrix}$

$$T^{-1}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}$$

If T^{-1} is linear (it is in fact) this tells us that its standard matrix

is $B = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$ Just check that $T^{-1}(v) = Bv$
is inverse of $T \rightarrow T^{-1}(T(v)) = B(Av)$
 $= (BA)v = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} v = v = \text{id}(v)$

★ Refer to next page!

which means that the equation's unique solution is $x = T^{-1}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 3/2 \end{bmatrix}$.

Similarly, $T^{-1}\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$ is the unique vector $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ such that $Ay = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

We again solve by row reduction to reduced echelon form:

$$\begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & -2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1/2 \end{bmatrix}$$

which means that $y = T^{-1}\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}$.

If we knew that T^{-1} were linear, then we could conclude that

$$T^{-1}(v) = Bv \quad \text{for} \quad B = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}.$$

This turns out to be the right formula for T^{-1} . To see why, just check that

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad BA = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

so $T \circ T^{-1} = T^{-1} \circ T = \text{id}_{\mathbb{R}^2}$.

It turns out that the inverse of an invertible linear transformation is always linear. Moreover, a linear transformation is invertible only if its standard matrix is square:

Proposition. If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear and invertible, then $n = m$ and T^{-1} is linear.

Proof. From results last time, we know that T is onto only if $n \geq m$ and one-to-one only if $n \leq m$. If T is both, then necessarily $n = m$.

Recall how T^{-1} is defined. For $u, v \in \mathbb{R}^n$ and $c \in \mathbb{R}$, we have

- $T^{-1}(u + v)$ = the unique vector x with $T(x) = u + v$.

Since $T(T^{-1}(u) + T^{-1}(v)) = T(T^{-1}(u)) + T(T^{-1}(v)) = u + v$, it follows that $x = T^{-1}(u) + T^{-1}(v)$.

- $T^{-1}(cv)$ = the unique vector y with $T(y) = cv$.

Since $T(cT^{-1}(v)) = cT(T^{-1}(v)) = cv$, it follows that $y = cT^{-1}(v)$.

These two items confirm that T^{-1} is linear. □

As usual, let's now translate the notion of invertibility for linear functions to matrices.

Definition. Let A be an $n \times n$ matrix and define $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $T(x) = Ax$. The matrix A is *invertible* if the function T is invertible. Its *inverse* is the unique matrix A^{-1} such that $T^{-1}(x) = A^{-1}x$.

This definition is natural enough, but a little abstract. Here is a more concrete formulation:

Proposition. Let A be an $n \times n$ matrix. The following mean the same thing:

- (1) A is invertible. *if A is standard matrix of an invertible linear function $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$.*
- (2) There is an $n \times n$ matrix A^{-1} such that $AA^{-1} = A^{-1}A = I_n$.

*Inverse matrix should have
size $n \times n$.*

$$I = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}$$

(3) For each $b \in \mathbb{R}^n$ the equation $Ax = b$ has a unique solution.

(4) $\text{RREF}(A) = I_n$.

Proof. To show that (1)-(4) are equivalent, we show that (1) implies (2), (2) implies (3), (3) implies (4), and (4) implies (1). This chain of implications shows that any of the properties implies any of the others.

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the linear transformation $T(X) = Ax$.

If A is invertible, then T is invertible with linear inverse $T^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

In this case if A^{-1} is the standard matrix of T^{-1} then

$$(T \circ T^{-1})(x) = A(A^{-1}x) = (AA^{-1})x = x$$

$$(T^{-1} \circ T)(x) = A^{-1}(Ax) = (A^{-1}A)x = x$$

for all $x \in \mathbb{R}^n$, which means that $AA^{-1} = A^{-1}A = I_n$. (Why?) Thus (1) $\Rightarrow$ (2).

If there is an $n \times n$ matrix A^{-1} such that $A^{-1}A = AA^{-1} = I_n$ then for each $b \in \mathbb{R}^n$, the unique solution to $Ax = b$ is $x = A^{-1}Ax = A^{-1}b$. (Check this!) Thus (2) $\Rightarrow$ (3).

If $Ax = b$ has a solution for every $b \in \mathbb{R}^n$ then A has a pivot position in every row. The solution to $Ax = b$ is unique if and only if $Ax = 0$ has only trivial solutions, which happens if and only if A has a pivot position in every column.

Thus if $Ax = b$ has a unique solution for every b then we must have $\text{RREF}(A) = I_n$, so (3) $\Rightarrow$ (4).

Finally, if $\text{RREF}(A) = I_n$ then A has a pivot position in every row and every column, so the columns of A are linearly independent (meaning T is one-to-one) and also span $\mathbb{R}^n$ (meaning T is onto), so T is invertible, which means A is invertible. So (4) $\Rightarrow$ (1). $\square$

A synonym for an invertible matrix is a *non-singular matrix*.

A matrix which is *not* invertible is sometimes called *singular*.

Example (Warning).

$$\text{Suppose } A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \text{ We have } AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

Neither A nor B is invertible, however.

The problem is there is no matrix A' such that $A'A = I_4$ and no matrix B' such that $BB' = I_3$.

A is the standard matrix of a linear transformation which is not one-to-one. (Why?)

B is the standard matrix of a linear transformation which is not onto. (Why?)

The following is a useful formula for small calculations.

Theorem. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be an arbitrary 2×2 matrix.

(1) A is invertible if and only if $ad - bc \neq 0$.

(2) If $ad - bc \neq 0$ then $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Proof. If $ad - bc = 0$ and either $b = 0$ or $d = 0$, then A has a row or column of all zeros (why?), so $\text{RREF}(A)$ is not the identity matrix and A is not invertible.

If $ad - bc = 0$ and $b \neq 0$ and $d \neq 0$, then A is row equivalent to

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \sim \begin{bmatrix} ad & bd \\ c & d \end{bmatrix} \sim \begin{bmatrix} ad & bd \\ -bc & -bd \end{bmatrix} \sim \begin{bmatrix} ad & bd \\ 0 & 0 \end{bmatrix}$$

so $\text{RREF}(A)$ cannot be the identity matrix, so A is not invertible.

If $ad - bc \neq 0$, then you can just check that $A^{-1}A = AA^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Try this yourself:

$$\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix} \quad \begin{matrix} da-bc & db-bd \\ -ca+ac & -cb+cd \end{matrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{ad-bc} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

□

Example. Recall in our earlier example we showed that $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$.

Some properties of the inverse of a matrix:

Theorem. Let A and B be $n \times n$ matrices.

1. If A is invertible then $(A^{-1})^{-1} = A$.
2. If A and B are both invertible then AB is invertible with $(AB)^{-1} = B^{-1}A^{-1}$.
3. If A is invertible then A^T is invertible with $(A^T)^{-1} = (A^{-1})^T$.

Proof.

1. To get a matrix C with $CA^{-1} = A^{-1}C = I_n$, take $C = A$.
2. Remember from last time that matrix multiplication is *associative*: this means that no matter how we parenthesize the product of a bunch of matrices, we get the same thing.
Use this as follows: if A and B are invertible then $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I_n$.
Likewise, $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I_n$. Therefore $(AB)^{-1} = B^{-1}A^{-1}$.
3. Observe that $A^T(A^{-1})^T = (A^{-1}A)^T = I_n^T = I_n$ and $(A^{-1})^T A^T = (AA^{-1})^T = I_n^T = I_n$.

□

A corollary of this theorem is that the product of a list of $n \times n$ invertible matrices is itself invertible, with inverse the product of the inverses in reverse order. In symbols: $(ABC \cdots Z)^{-1} = Z^{-1} \cdots C^{-1}B^{-1}A^{-1}$.

Checking whether an $n \times n$ matrix is invertible is nearly the same process as computing its inverse.

Process to compute A^{-1}

Let A be an $n \times n$ matrix. Consider the $n \times 2n$ matrix $\begin{bmatrix} A & I_n \end{bmatrix}$.

If A is invertible then $\text{RREF} \left(\begin{bmatrix} A & I_n \end{bmatrix} \right) = \begin{bmatrix} I_n & A^{-1} \end{bmatrix}$.

So to compute A^{-1} , row reduce $\begin{bmatrix} A & I_n \end{bmatrix}$ to reduced echelon form, and then take the last n columns.

Example. To find the inverse of $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$ we row reduce

$$\begin{aligned} \left[\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right] &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -9/2 & 7 & -3/2 \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -9/2 & 7 & -3/2 \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{array} \right]. \end{aligned}$$

Now check directly that $A^{-1} = \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix}!$

For any $n \times n$ matrix A ,

$$\text{RREF} \left(\begin{bmatrix} A & I \end{bmatrix} \right) = \begin{bmatrix} \text{RREF}(A) & B \end{bmatrix}$$

$n \times 2n$ matrix for some $n \times n$ B .

Fact: If $\text{RREF}(A) = I$ then $A^{-1} = B$
 If $\text{RREF}(A) \neq I$ then A^{-1} does not exist.

Note: $\text{RREF}([I \ A]) = [I \ A]$

After computing B , you should check that $AB = I$
 $BA = I$

Q Which 1×1 matrices $A = [a]$ are invertible?

We need to find $A^{-1} = [b]$ with $AA^{-1} = [a][b] = [ab] = [1]$

$$A^{-1}A = [b][a] = [ba] = 1$$

we want $b = \frac{1}{a}$ (only exists if $a \neq 0$)

the ones with $a \neq 0 : [a]^{-1} = [\frac{1}{a}]$.

4 Vocabulary

Keywords from today's lecture:

1. Identity matrix $I = I_n$.

The $n \times n$ matrix that has 1 in every diagonal position and 0 in every other position.

If A is an $m \times n$ matrix then $I_m A = A I_n = A$.

If $v \in \mathbb{R}^n$ then $I_n v = v$ (since v is an $n \times 1$ matrix).

We have $I_1 = \begin{bmatrix} 1 \end{bmatrix}$, $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, and $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

2. Elementary matrix.

A matrix obtained by doing one row operation to the $n \times n$ identity matrix.

Examples when $n = 3$ include $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, and $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

3. Identity function $\text{id}_X : X \rightarrow X$.

The function with $\text{id}_X(x) = x$ for all $x \in X$.

If $X = \mathbb{R}^4$ then id_X is the linear transformation with standard matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

4. Invertible function.

A function $f : X \rightarrow Y$ such that for each $y \in Y$ there is exactly one element $x \in X$ with $f(x) = y$.

This occurs precisely when there is an *inverse function* $g : Y \rightarrow X$ such that $g(f(x)) = x$ and $f(g(y)) = y$ for all $x \in X$ and $y \in Y$, that is, with $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$.

5. Invertible matrix and the *inverse* of a matrix.

A matrix is invertible if it is square and is the standard matrix of an invertible linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^n$ for some n .

An $n \times n$ matrix A is invertible precisely when $\text{RREF}(A) = I_n$ is the $n \times n$ identity matrix.

If A is invertible, then there exists a unique matrix A^{-1} such that $AA^{-1} = A^{-1}A = I_n$.

The matrix A^{-1} is called the inverse of A .

If A is $n \times n$ and invertible then $\text{RREF}(\begin{bmatrix} A & I_n \end{bmatrix}) = \begin{bmatrix} I_n & A^{-1} \end{bmatrix}$.

Row reducing gives the most efficient way of calculating A^{-1} .

Example: If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then A is invertible if and only if $ad - bc \neq 0$.

When this happens, $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.